

STA 013 — Problem Set (Day 4)

Topic: Conditional Probability, Independence, Random Variables | Day 4 (Professor Max)

Instructions: For each question, first identify what operation is being asked for (union, intersection, complement, conditional), then compute. Show your work for calculation questions.

1. Match each phrase to the correct probability operation:

- (i) "What is the probability that neither A nor B occurs?"
 - (ii) "What is the probability that at least one of A or B occurs?"
 - (iii) "Given that B has occurred, what is the probability of A?"
 - (iv) "What is the probability that both A and B occur?"
- A) (i) complement; (ii) union; (iii) conditional; (iv) intersection
B) (i) intersection; (ii) complement; (iii) union; (iv) conditional
C) (i) union; (ii) intersection; (iii) complement; (iv) conditional
D) (i) conditional; (ii) union; (iii) complement; (iv) intersection

2. A study of 100 patients classifies them by smoking status and lung disease:

	Lung Disease	No Disease	Total
Smoker	40	10	50
Non-Smoker	10	40	50
Total	50	50	100

What is $P(\text{Disease} \mid \text{Smoker})$, and what does it mean in plain language?

- A) $40/100 = 0.40$ — 40% of all patients have lung disease and smoke
 - B) $40/50 = 0.80$ — given that a patient smokes, there is an 80% chance they have lung disease
 - C) $50/100 = 0.50$ — half the population smokes
 - D) $10/50 = 0.20$ — given that a patient has no disease, there is a 20% chance they smoke
- Hint: The condition is "Smoker" — so your denominator is the Smoker row total, not the grand total.

3. Suppose $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cap B) = 0.12$. Are A and B independent?

- A) Yes, because $P(A) + P(B) = 0.7$ which is less than 1
- B) Yes, because $P(A) \times P(B) = 0.4 \times 0.3 = 0.12 = P(A \cap B)$
- C) No, because $P(A \cap B)$ should equal 0 for independence
- D) No, because $P(A) \neq P(B)$

4. You are told that $P(A \mid B) = 0.5$. What is $P(A \mid B^c)$?

- A) 0.5, since complementing the condition doesn't change the probability of A

- B) 0.5, since $P(A | B^c) = 1 - P(A | B) = 1 - 0.5 = 0.5$ (coincidence in this case)
- C) Cannot be determined from the given information alone — complementing the condition changes the world you're computing within
- D) Always equal to $P(A | B)$ for any events A and B

→ Hint: $P(A^c | B) = 1 - P(A | B)$ is TRUE. But $P(A | B^c) = 1 - P(A | B)$ is NOT true in general.

5. Events A and B are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.4$. Are A and B independent?

- A) Yes, because mutually exclusive events are always independent
- B) Yes, because $P(A) + P(B) < 1$
- C) No — mutually exclusive events with positive probability are actually dependent: $P(A \cap B) = 0$, but $P(A) \times P(B) = 0.12 \neq 0$
- D) Cannot be determined without knowing $P(A \cap B)$

→ Hint: For independence, $P(A \cap B)$ must equal $P(A) \times P(B)$. Mutually exclusive means $P(A \cap B) = 0$. Are those the same?

6. Using the heart attack population from lecture ($P(\text{high risk}) = 0.10$), three people are randomly selected. What is the probability that EXACTLY TWO of the three are high risk?

- A) 0.081
- B) 0.243
- C) 0.027
- D) 0.001

→ Hint: List the 3 arrangements with exactly 2 high-risk (HHN, HNH, NHH), compute each probability, and add them up.

7. In a population that is 49% female, 8% of females are high risk for heart disease. A single person is selected at random. What is $P(\text{high risk} \cap \text{female})$?

- A) 0.08, since that's the probability of high risk given female
- B) 0.49, since that's the probability of being female
- C) 0.0392, computed as $P(H | F) \times P(F) = 0.08 \times 0.49$
- D) 0.57, computed as $P(H | F) + P(F)$

→ Hint: "High risk AND female" is an intersection. Use $P(H \cap F) = P(H | F) \times P(F)$, choosing the version where the values given plug in directly.

8. Roll two fair dice. Using the probability distribution table from lecture, what is $P(\text{sum} \geq 10)$?

- A) 3/36
- B) 1/12
- C) 1/6
- D) 1/4

→ Hint: "At least 10" includes sums of 10, 11, and 12. Add those probabilities: $3/36 + 2/36 + 1/36$.

9. Roll two fair dice. Given that the sum is ODD, what is the probability that the sum is 7?

- A) $6/36 = 1/6$
- B) $6/18 = 1/3$
- C) $1/11$
- D) $1/2$

→ Hint: $P(\text{sum}=7 \mid \text{sum is odd}) = P(\text{sum}=7 \cap \text{odd}) / P(\text{odd})$. The odd sums are 3,5,7,9,11 — count their total outcomes.

10. For two fair dice, are the events “sum = 7” and “sum > 9” independent, mutually exclusive, both, or neither?

- A) Independent, because they involve different sums
- B) Mutually exclusive only — a sum cannot be both exactly 7 and greater than 9 simultaneously, and this also means they are NOT independent
- C) Both independent and mutually exclusive
- D) Neither — they can occur at the same time

→ Hint: Can a single roll produce both outcomes? Then check: does $P(A \cap B) = P(A) \times P(B)$?

11. $P(A \mid B) = 0.7$. What is $P(A^c \mid B)$?

- A) 0.7
- B) 0.3
- C) Cannot be determined
- D) 1.7

→ Hint: This is complementing the EVENT (A), not the condition (B). The rule $P(A^c \mid B) = 1 - P(A \mid B)$ DOES apply here.

12. A survey of 100 people is summarized below. Based on the table, are events A and B independent?

	B	B ^c	Total
A	15	35	50
A ^c	15	35	50
Total	30	70	100

- A) No, because A and B appear in the same table
- B) No, because $P(A) \neq P(B)$
- C) Yes — $P(A) \times P(B) = 0.5 \times 0.3 = 0.15 = P(A \cap B) = 15/100$
- D) Cannot be determined from a table alone

→ Hint: Compute $P(A)$, $P(B)$, and $P(A \cap B)$ from the table. Then check whether $P(A) \times P(B) = P(A \cap B)$.

Answer Key & Explanations

- 1. A** — "Neither A nor B" = complement of $(A \cup B)$; "at least one of A or B" = union; "given B" = conditional; "both A and B" = intersection.
- 2. B** — $P(\text{Disease} \mid \text{Smoker}) = (\text{smokers with disease}) / (\text{all smokers}) = 40/50 = 0.80$. The condition "Smoker" makes the smoker row total (50) the denominator, not the grand total.
- 3. B** — Independence check: $P(A) \times P(B) = 0.4 \times 0.3 = 0.12 = P(A \cap B)$. Since the product of their individual probabilities equals their joint probability, A and B are independent.
- 4. C** — $P(A \mid B^c)$ cannot be determined from $P(A \mid B)$ alone. Complementing the CONDITION changes the entire reference world you're computing within — the rule $1 - P(A \mid B)$ only applies when you complement the EVENT (giving $P(A^c \mid B)$), not the condition.
- 5. C** — Mutually exclusive means $P(A \cap B) = 0$. But $P(A) \times P(B) = 0.3 \times 0.4 = 0.12 \neq 0$. Since $P(A \cap B) \neq P(A) \times P(B)$, they are NOT independent. In fact, mutually exclusive events with positive probability are always dependent.
- 6. C** — Three arrangements with exactly 2 high-risk: HHN, HNH, NHH. Each has probability $(0.1)(0.1)(0.9) = 0.009$. Total: $3 \times 0.009 = 0.027$.
- 7. C** — $P(H \cap F) = P(H \mid F) \times P(F) = 0.08 \times 0.49 = 0.0392$. Use this version because $P(F) = 0.49$ is directly given.
- 8. C** — $P(\text{sum} \geq 10) = P(10) + P(11) + P(12) = 3/36 + 2/36 + 1/36 = 6/36 = 1/6$.
- 9. B** — Odd sums (3,5,7,9,11) account for $2+4+6+4+2 = 18$ of 36 outcomes. $P(\text{odd}) = 18/36$. $P(\text{sum}=7 \text{ and odd}) = P(\text{sum}=7) = 6/36$. $P(7 \mid \text{odd}) = (6/36)/(18/36) = 6/18 = 1/3$.
- 10. B** — A single roll can't be both 7 and >9 , so $P(A \cap B) = 0$ — mutually exclusive. But $P(A) \times P(B) = (6/36) \times (6/36) = 1/36 \neq 0$, so they are NOT independent. Mutually exclusive events with positive probability are never independent.
- 11. B** — $P(A^c \mid B) = 1 - P(A \mid B) = 1 - 0.7 = 0.3$. Complementing the EVENT (not the condition) within the same conditional world follows the standard complement rule.
- 12. C** — $P(A) = 50/100 = 0.5$, $P(B) = 30/100 = 0.3$, $P(A \cap B) = 15/100 = 0.15$. Since $0.5 \times 0.3 = 0.15 = P(A \cap B)$, A and B are independent. Notice also that the rows are proportional (15:35 in both A and A^c) — a visual shortcut for spotting independence in a 2×2 table.